Accelerated Algebra and Analysis: Algebra Part

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Abstract

This is a simple abstract for the Accelerated Algebra and Analysis notes it is focused on the algebra part.

1 Lecture 1

Vectors, Matrices, and Complex numbers

1.1 Geometric motivation

Cubic curves are plain curves given by

$$f(x, y) = 0 \quad \text{where } f \text{ is a cubic polynomial in } x \text{ and } y$$

Theorem 1.1

Chasles's theorem

Let C_3 and C_3' be two cubic curves such that their intersection $C_3 \cap C_3'$ consists of 9 points. Let Z_3 be the set of 8 intersection points

$$Z_3 = (C_3 \cap C_3') \setminus \{P\}$$

where P is the ninth intersection point. Then any cubic curve passing through the points of Z_3 also passes through P.

Chasles's theorem implies Pascal's theorem.

Theorem 1.2

Pascal's theorem

Let A, B, C, D, E, F be the six vertices of a hexagon inscribed in a conic. Then the points of intersection of the pairs of opposite sides $(AB \cap DE), (BC \cap EF), (CD \cap FA)$ are collinear.

- Hexagrammum Mysticum.
- Pappus line

1.2 Vectors

1.2.1 Multidimensional vectors

An n-dimensional vector is an ordered list with n components. It can be represented as

$$\mathbf{v} = (v_1, v_2, \ldots, v_n)$$

We usually write this vector as a column vector

$$\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix}$$

Notation

In handwritten notes, an n-dimensional vector $\mathbf{x}$ is often denoted by

$$\overline{x} \quad \underline{x} \quad \mathbf{x} \quad \vec{x}$$

Notation

The set of all n-dimensional vectors is denoted by $\mathbb{R}^n$

$$\mathbb{R}^n = \{(v_1, v_2, \ldots, v_n) \mid v_i \in \mathbb{R} \text{ for } i = 1, 2, \ldots, n\}$$

A special vector in $\mathbb{R}^n$ is the zero vector

$$\mathbf{0} = (0, 0, \ldots, 0)$$

1.2.2 Operations with vectors

Let $\mathbf{x} = (x_1, x_2, \ldots, x_n)$ and $\mathbf{y} = (y_1, y_2, \ldots, y_n)$ be two vectors in $\mathbb{R}^n$.

- **Vector addition:**

$$\mathbf{x} + \mathbf{y} = \begin{pmatrix} x_1 + y_1 \\ x_2 + y_2 \\ \vdots \\ x_n + y_n \end{pmatrix}$$

- **Scalar multiplication:**

$$c\mathbf{x} = \begin{pmatrix} cx_1 \\ cx_2 \\ \vdots \\ cx_n \end{pmatrix} \quad \text{for } c \in \mathbb{R}$$

Let $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^n$. The basic properties of vector operations are

1. $\mathbf{x} + \mathbf{y} = \mathbf{y} + \mathbf{x}$
2. $(\mathbf{x} + \mathbf{y}) + \mathbf{z} = \mathbf{x} + (\mathbf{y} + \mathbf{z})$

3. $c(\mathbf{x}+\mathbf{y}) = c\mathbf{x} + c\mathbf{y}$ for $c \in \mathbb{R}$
4. $(cd)\mathbf{x} = c(d\mathbf{x})$ for $c, d \in \mathbb{R}$
5. $\mathbf{x} + \mathbf{0} = \mathbf{x}$
6. $\mathbf{x} + (-\mathbf{x}) = \mathbf{0}$
7. $1\mathbf{x} = \mathbf{x}$ for all $\mathbf{x} \in \mathbb{R}^n$
8. $0\mathbf{x} = \mathbf{0}$ for all $\mathbf{x} \in \mathbb{R}^n$

1.2.3 Length and distance

Definition 1.1

The length (or norm) of a vector $\mathbf{x} \in \mathbb{R}^n$ is defined as

$$\|\mathbf{x}\| = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}$$

Definition 1.2

The distance between two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ is defined as

$$d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|$$

Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$. Then the distance between $\mathbf{x}$ and $\mathbf{y}$ satisfies the following properties

1. $d(\mathbf{x}, \mathbf{y}) \geq 0$ and $d(\mathbf{x}, \mathbf{y}) = 0$ if and only if $\mathbf{x} = \mathbf{y}$
2. $d(\mathbf{x}, \mathbf{y}) = d(\mathbf{y}, \mathbf{x})$
3. $d(\mathbf{x}, \mathbf{z}) \leq d(\mathbf{x}, \mathbf{y}) + d(\mathbf{y}, \mathbf{z})$ (triangle inequality)
4. $d(a\mathbf{x}, a\mathbf{y}) = |a| d(\mathbf{x}, \mathbf{y})$ for $a \in \mathbb{R}$

1.2.4 Dot product

Definition 1.3

Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^3$. The dot product of $\mathbf{x}$ and $\mathbf{y}$ is defined by

$$\mathbf{x} \cdot \mathbf{y} = x_1 y_1 + x_2 y_2 + x_3 y_3$$

Let $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^n$ and $c \in \mathbb{R}$. The dot product satisfies the following properties

1. $\mathbf{x} \cdot \mathbf{y} = \mathbf{y} \cdot \mathbf{x}$ (commutativity)
2. $(\mathbf{x} + \mathbf{y}) \cdot \mathbf{z} = \mathbf{x} \cdot \mathbf{z} + \mathbf{y} \cdot \mathbf{z}$ (distributivity)
3. $(c\mathbf{x}) \cdot \mathbf{y} = c(\mathbf{x} \cdot \mathbf{y})$ (scalar multiplication)
4. $\mathbf{x} \cdot \mathbf{x} = \|\mathbf{x}\|^2$
5. $\mathbf{x} \cdot \mathbf{0} = 0$

- If $\mathbf{x} \cdot \mathbf{y} = 0$, we call $\mathbf{x}$ and $\mathbf{y}$ orthogonal
- If $|\mathbf{x} \cdot \mathbf{y}| = \|\mathbf{x}\|\|\mathbf{y}\|$, they are collinear

Since

$$\mathbf{x} \cdot \mathbf{y} = \|\mathbf{x}\|\|\mathbf{y}\| \cos\theta$$

1.2.5 Cauchy-Schwarz inequality

Theorem 1.3

Cauchy-Schwarz inequality

Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$. The Cauchy-Schwarz inequality states that

$$|\mathbf{x} \cdot \mathbf{y}| \leq \|\mathbf{x}\|\|\mathbf{y}\|$$

with equality if and only if $\mathbf{x}$ and $\mathbf{y}$ are linearly dependent.

Proof. Consider the function

$$f(\lambda) = \|\mathbf{x} + \lambda\mathbf{y}\|^2$$

Since the norm is always non-negative, we have $f(\lambda) \geq 0$ for all $\lambda \in \mathbb{R}$.

$$\|\mathbf{x} + \lambda\mathbf{y}\|^2 = (\mathbf{x} + \lambda\mathbf{y}) \cdot (\mathbf{x} + \lambda\mathbf{y}) = \mathbf{x} \cdot \mathbf{x} + 2\lambda(\mathbf{x} \cdot \mathbf{y}) + \lambda^2(\mathbf{y} \cdot \mathbf{y}) \geq 0$$

This is a quadratic inequality in λ.
For it to hold for all $\lambda \in \mathbb{R}$, the discriminant must be non-positive:

$$(2(\mathbf{x} \cdot \mathbf{y}))^2 - 4(\mathbf{y} \cdot \mathbf{y})(\mathbf{x} \cdot \mathbf{x}) \leq 0$$

Simplifying, we get

$$(\mathbf{x} \cdot \mathbf{y})^2 \leq (\mathbf{x} \cdot \mathbf{x})(\mathbf{y} \cdot \mathbf{y}) = \|\mathbf{x}\|^2\|\mathbf{y}\|^2$$

which is the Cauchy-Schwarz inequality. □

1.2.6 Triangular inequality

Theorem 1.4

Triangular inequality

Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$. The triangular inequality states that

$$\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$$

with equality if and only if one vector is a non-negative scalar multiple of the other.

Proof. Consider the square of the norm of the sum:

$$\|\mathbf{x} + \mathbf{y}\|^2 = (\mathbf{x} + \mathbf{y}) \cdot (\mathbf{x} + \mathbf{y}) = \|\mathbf{x}\|^2 + 2(\mathbf{x} \cdot \mathbf{y}) + \|\mathbf{y}\|^2$$

Using the Cauchy-Schwarz inequality, we have

$$2(\mathbf{x} \cdot \mathbf{y}) \leq 2\|\mathbf{x}\|\|\mathbf{y}\|$$

Therefore

$$\|\mathbf{x} + \mathbf{y}\|^2 \leq \|\mathbf{x}\|^2 + 2\|\mathbf{x}\|\|\mathbf{y}\| + \|\mathbf{y}\|^2 = (\|\mathbf{x}\| + \|\mathbf{y}\|)^2$$

Taking the square root of both sides gives the triangular inequality

$$\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$$

□

1.2.7 Angles between vectors

Definition 1.4

The angle θ between two non-zero vectors $\mathbf{x}$ and $\mathbf{y}$ is defined by the formula

$$\cos\theta = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\|\|\mathbf{y}\|}$$

Lemma 1.5

Pythagoras theorem for vectors states that if two vectors $\mathbf{x}$ and $\mathbf{y}$ are orthogonal, then

$$\|\mathbf{x} + \mathbf{y}\|^2 = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2$$

If $\mathbf{x}$ and $\mathbf{y}$ are orthogonal (perpendicular), then

$$\mathbf{x} \cdot \mathbf{y} = 0$$

and

$$\theta = \frac{\pi}{2}$$

1.2.8 Cross product of three-dimensional vectors

Definition 1.5

Let $\mathbf{x} = (x_1, x_2, x_3)$ and $\mathbf{y} = (y_1, y_2, y_3)$ be vectors in $\mathbb{R}^3$.
The cross product $\mathbf{x} \times \mathbf{y}$ is defined as

$$\mathbf{x} \times \mathbf{y} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{vmatrix}$$

where $\mathbf{i}, \mathbf{j}, \mathbf{k}$ are the standard unit vectors in $\mathbb{R}^3$.

$$\mathbf{x} \times \mathbf{y} = \begin{pmatrix} x_2y_3 - x_3y_2 \\ -x_1y_3 + x_3y_1 \\ x_1y_2 - x_2y_1 \end{pmatrix}$$

The cross product has the following properties:

1. $\mathbf{x} \times \mathbf{y} = -(\mathbf{y} \times \mathbf{x})$
2. $\mathbf{x} \times (\mathbf{y} + \mathbf{z}) = \mathbf{x} \times \mathbf{y} + \mathbf{x} \times \mathbf{z}$
3. $(a\mathbf{x}) \times \mathbf{y} = a(\mathbf{x} \times \mathbf{y})$ for $a \in \mathbb{R}$
4. $\|\mathbf{x} \times \mathbf{x}\| = 0$
5. $(\mathbf{x} \times \mathbf{y}) \cdot \mathbf{x} = 0$ and $(\mathbf{x} \times \mathbf{y}) \cdot \mathbf{y} = 0$
6. $\|\mathbf{x} \times \mathbf{y}\| = \|\mathbf{x}\|\|\mathbf{y}\| \sin\theta$, where θ is the angle between $\mathbf{x}$ and $\mathbf{y}$

1.3 Matrix

1.3.1 Definition

Definition 1.6

A matrix is a rectangular array of numbers arranged in rows and columns. An $m \times n$ matrix has m rows and n columns and is typically denoted by

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}$$

Zero matrix is a matrix in which all the elements are zero, typically denoted by

$$0 = \begin{pmatrix} 0 & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 \end{pmatrix}$$

$n \times n$ identity matrix is a matrix in which all the elements on the main diagonal are 1 and all other elements are 0, typically denoted by

$$I_n = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}$$

1.3.2 Sum representation

Any $m \times n$ matrix $A = (a_{ij})$ can be represented as the sum of its elements multiplied by the corresponding standard basis matrices

$$A = \sum_{i=1}^{m} \sum_{j=1}^{n} a_{ij} E_{ij},$$

where E_{ij} is the $m \times n$ matrix with 1 in the (i, j)-th position and 0 elsewhere.

1.3.3 Algebra operations with matrices

$A + B$

$$(A + B)_{ij} = A_{ij} + B_{ij}$$

cA

$$(cA)_{ij} = c \cdot A_{ij}$$

AB

$$(AB)_{ij} = \sum_{k=1}^{n} A_{ik} B_{kj}$$

1.3.4 Matrix multiplication

Definition 1.7

matrix multiplication

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix} \quad B = \begin{pmatrix} b_{11} & b_{12} & \dots & b_{1q} \\ b_{21} & b_{22} & \dots & b_{2q} \\ \vdots & \vdots & \ddots & \vdots \\ b_{p1} & b_{p2} & \dots & b_{pq} \end{pmatrix}$$

$$AB = \begin{pmatrix} (AB)_{11} & (AB)_{12} & \dots & (AB)_{1q} \\ (AB)_{21} & (AB)_{22} & \dots & (AB)_{2q} \\ \vdots & \vdots & \ddots & \vdots \\ (AB)_{m1} & (AB)_{m2} & \dots & (AB)_{mq} \end{pmatrix}$$

Make sure that the number of columns of A matches the number of rows of B for the matrix multiplication AB to be defined,

Example

If A is an $m \times n$ matrix and B is a $p \times q$ matrix, then $n = p$. In particular, the product AB will be an $m \times q$ matrix.

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}_{m \times n}, \quad B = \begin{pmatrix} b_{11} & b_{12} & \dots & b_{1q} \\ b_{21} & b_{22} & \dots & b_{2q} \\ \vdots & \vdots & \ddots & \vdots \\ b_{p1} & b_{p2} & \dots & b_{pq} \end{pmatrix}_{p \times q}$$

$$AB = \begin{pmatrix} (AB)_{11} & (AB)_{12} & \dots & (AB)_{1q} \\ (AB)_{21} & (AB)_{22} & \dots & (AB)_{2q} \\ \vdots & \vdots & \ddots & \vdots \\ (AB)_{m1} & (AB)_{m2} & \dots & (AB)_{mq} \end{pmatrix}_{m \times q}$$

Basic properties of matrix addition and scalar multiplication

1. $(A + B) = (B + A)$

2. $(A+B)+C = A+(B+C)$
3. $A+0 = A$
4. $c(A+B) = cA+cB, \quad c \in \mathbb{R}$

Basic properties of matrix multiplication

1. $(AB)C = A(BC)$
2. $A(B+C) = AB+AC$
3. $(A+B)C = AC+BC$
4. $AI_n = A = I_m A_{m\times n}$

For $(AB)C = A(BC)$

Proof. Let A be an $m \times n$ matrix, B be an $n \times p$ matrix, and C be a $p \times q$ matrix. Then

$$\begin{aligned}((AB)C)_{ij} &= \sum_{k=1}^{p}(AB)_{ik}C_{kj} \\ &= \sum_{k=1}^{p}\left(\sum_{l=1}^{n}A_{il}B_{lk}\right)C_{kj} \\ &= \sum_{l=1}^{n}A_{il}\left(\sum_{k=1}^{p}B_{lk}C_{kj}\right) \\ &= (A(BC))_{ij}, \quad 1 \le i \le m, \quad 1 \le j \le q\end{aligned}$$

Therefore, $(AB)C = A(BC)$. □

1.3.5 Column Vectors

Definition 1.8

Column vectors $(x_1, x_2, \ldots, x_m)^T$

A column vector is an $m \times 1$ matrix, i.e.,

$$\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_m \end{pmatrix}_{m\times 1}$$

If

$$A = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{pmatrix}$$

Then

$$MA = \begin{pmatrix} (MA)_1 \\ (MA)_2 \\ \vdots \\ (MA)_m \end{pmatrix}$$

1.3.6 Invertible matrices

Definition 1.9

An $n \times n$ matrix A is said to be invertible if there exists an $n \times n$ matrix B such that

$$AB = BA = I_n.$$

The matrix B is called the inverse of A and is denoted by A^{-1}.

A matrix is called singular if it is an $n \times n$ matrix that is not invertible.

Notation

The set of all invertible $n \times n$ matrices is denoted by $GL(n, \mathbb{R})$ if the entries are real numbers.

For 2×2 case, an invertible matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ has an inverse given by

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \quad ad - bc \neq 0.$$

1.3.7 Transpose

Definition 1.10

Let A be an $m \times n$ matrix,

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}_{m \times n}.$$

The transpose of A, denoted by A^T, is the $n \times m$ matrix obtained by interchanging the rows and columns of A:

$$A^T = \begin{pmatrix} a_{11} & a_{21} & \dots & a_{m1} \\ a_{12} & a_{22} & \dots & a_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \dots & a_{mn} \end{pmatrix}_{n \times m}.$$

Properties of the transpose

1. $(A^T)^T = A$

2. $(A+B)^T = A^T + B^T$
3. $(cA)^T = cA^T$ for any scalar c
4. $(AB)^T = B^T A^T$

Definition 1.11

Symmetric matrix A matrix A is called symmetric if $A^T = A$.

Skew-symmetric matrix A matrix A is called skew-symmetric if $A^T = -A$.

Theorem 1.6

For any two $n \times n$ matrices A and B, the matrix $AB - BA$ is skew-symmetric.

Proof. We need to show that $(AB - BA)^T = -(AB - BA)$.

$$\begin{aligned}(AB - BA)^T &= (AB)^T - (BA)^T \\ &= B^T A^T - A^T B^T \\ &= -(A^T B^T - B^T A^T) \\ &= -(AB - BA).\end{aligned}$$

Hence, $AB - BA$ is skew-symmetric. □

1.4 Complex Number

1.4.1 Definition

We adjoin the imaginary unit i in $\mathbb{R}$, where

$$i^2 = -1$$

We define a complex number as

$$z = a + bi, \quad a, b \in \mathbb{R}.$$

And we denote the set of all complex numbers by $\mathbb{C}$

$$\mathbb{C} = \{a + bi \mid a, b \in \mathbb{R}\}$$

1.4.2 Fundamental theorem of Algebra

Theorem 1.7

Let

$$p(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_1 z + a_0$$

be a non-constant polynomial with complex coefficients.
If $n \geq 1$, then there exists at least one complex number $z_0 \in \mathbb{C}$ such that

$$p(z_0) = 0.$$

In other words, every non-constant polynomial with complex coefficients has at least one root in $\mathbb{C}$.

Lemma 1.8

Let

$$p(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_1 z + a_0$$

be a non-constant polynomial with complex coefficients.
Then $p(z)$ can be factored into linear factors over $\mathbb{C}$, i.e.,

$$p(z) = a_n (z - z_1)(z - z_2) \dots (z - z_n),$$

where $z_1, z_2, \dots, z_n \in \mathbb{C}$ are the roots of $p(z)$.

1.5 Field

1.5.1 Definition

Definition 1.12

A field is a set F equipped with addition and multiplication, such that the following properties hold

1. F is an abelian group under addition, with identity element 0.
2. $F \setminus \{0\}$ is an abelian group under multiplication, with identity element 1.
3. Multiplication is distributive over addition, i.e., for all $a, b, c \in F$,

$$a(b + c) = ab + ac.$$

Definition 1.13

Abelian group

A group G satisfies the commutative property if for all $a, b \in G$

$$a + b = b + a$$

1.5.2 Other finite fields

To construct many finite fields, fix a prime number $p \geq 2$. Let $\mathbb{F}_p$ be the following set consisting of p integers:

$$\{0, 1, \dots, p - 1\}.$$

Equip $\mathbb{F}_p$ with operations $+$ and $*$ as follows:

1. For every two elements a and b in $\mathbb{F}_p$, let $a + b =$ the remainder of the integer $a + b$ when divided by p.
2. For every two elements a and b in $\mathbb{F}_p$, let $a * b =$ the remainder of the integer $a * b$ when divided by p.

Then $\mathbb{F}_p$ equipped with $+$ and $*$ is a field.

Example

Suppose that $p = 1973$ (it is prime). Then

$$\frac{a}{45} = a * 570$$

for every a in $\mathbb{F}_p$, since $45 * 570 = 1$ in $\mathbb{F}_p$.

1.5.3 Solving system of two linear equations

Example

Let us solve the following system of equations in $\mathbb{F}_7$:

$$\begin{cases} 2x + 3y = 5, \\ 3x + 4y = 1. \end{cases}$$

Observe that $2 \times 4 = 1$ in $\mathbb{F}_7$. Then $\frac{1}{2} = 4$ and

$$2x + 3y = 5 \implies x = 4(5 - 3y),$$

which gives $x = 6 + 2y$. Then

$$3(6 + 2y) + 4y = 1,$$

so that $3y = 4$, which gives $y = 5 \times 4 = 6$. Then

$$(x, y) = (6 + 2 \times 6, 6) = (4, 6).$$

1.5.4 A better way to construct a field with 4 elements

Start with the field $\mathbb{F}_2$ consisting of two elements 0 and 1.
Introduce a new element t such that

$$t^2 = t + 1$$

Consider all sums (there are 4 of them) that look like

$$a + bt$$

where a and b are both in $\mathbb{F}_2$. Denote the obtained set by $\mathbb{F}_4$. Then extend $+$ and $*$ from $\mathbb{F}_2$ to elements in $\mathbb{F}_4$ as follows:

$$(a + bt) * (c + dt) = ac + bdt^2 + adt + bct = (ac + bd) + (ad + bc + bd)t$$

and

$$(a + bt) + (c + dt) = (a + c) + (b + d)t$$

for any elements a, b, c, d in the field $\mathbb{F}_2$. Then $\mathbb{F}_4 = \{0, 1, t, 1 + t\}$ is a field.

2 Lecture 2

Linear subspaces and their bases

2.1 Linear Subspaces

2.1.1 Hyperplanes

Definition 2.1

A hyperplane in $\mathbb{R}^n$ is a set of points x in $\mathbb{R}^n$ that satisfy a linear equation of the form

$$\sum_{i=1}^{n} A_i x_i = B,$$

where $A_1, A_2, \ldots, A_n$ are not all zero, and $A_1, \ldots, A_n, B$ are in $\mathbb{R}$.

Remark

A hyperplane contains the origin if and only if $B = 0$.

2.1.2 Linear subspaces

Definition 2.2

A linear subspace (or vector subspace) of $\mathbb{R}^n$ is a set of points x in $\mathbb{R}^n$ such that

- if x and y are in the subspace V, then

$$x + y \in V.$$

- if x is in the subspace and α is a scalar in $\mathbb{R}$, then

$$\alpha x \in V.$$

- the zero vector 0 is in the subspace.

$$0 \in V.$$

Notation

Linear subspace = Vector subspace

Example

- $\{0\}$ is a linear subspace of $\mathbb{R}^n$, which is trivial linear subspace.
- a line through the origin in $\mathbb{R}^n$ is also a linear subspace.
- a hyperplane through the origin in $\mathbb{R}^n$ is also a linear subspace.
- whole $\mathbb{R}^n$ is also a linear subspace.

2.1.3 Kernel

Definition 2.3

Kernel of a matrix A is the set of all vectors x such that $Ax = 0$.

$$\ker(A) = \{x \in \mathbb{R}^n \mid Ax = 0\}$$

the kernel is a linear subspace of $\mathbb{R}^n$.

Notation

Some people called the kernel of a matrix A the null space of A.

Proof. To prove that the kernel is a linear subspace, we need to show that it satisfies the three conditions for a linear subspace

- If x and y are in $\ker(A)$, then $Ax = 0$ and $Ay = 0$.

  $$A(x + y) = Ax + Ay = 0 + 0 = 0,$$

  which means $x + y \in \ker(A)$.

- If x is in $\ker(A)$ and α is a scalar in $\mathbb{R}$, then $Ax = 0$.

  $$A(\alpha x) = \alpha Ax = \alpha \cdot 0 = 0,$$

  which means $\alpha x \in \ker(A)$.

- The zero vector 0 is in $\ker(A)$ because

  $$A0 = 0.$$

□

2.2 Linear Combinations and Span

2.2.1 Linear Combination

Definition 2.4

A linear combination of vectors $v_1, v_2, \ldots, v_k$ in $\mathbb{R}^n$ is a vector of the form

$$v = \alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_k v_k,$$

where $\alpha_1, \alpha_2, \ldots, \alpha_k$ are scalars in $\mathbb{R}$.

Example

Matrix multiplication can be seen as a linear combination of the columns of the matrix. For example, if A is an $m \times n$ matrix with columns $A_1, A_2, \ldots, A_n$ and x is a vector in $\mathbb{R}^n$ with components $x_1, x_2, \ldots, x_n$, then

$$Ax = x_1 A_1 + x_2 A_2 + \cdots + x_n A_n.$$

Let V contains list of vectors $v_1, v_2, \ldots, v_k$ in $\mathbb{R}^n$. Then it contains all linear combinations of these vectors, i.e.,

$$\{\alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_k v_k \mid \alpha_1, \alpha_2, \ldots, \alpha_k \in \mathbb{R}\} \subseteq V.$$

Remark

The zero is a linear combination for all sets of vectors, obtained by taking all the scalars $\alpha_1, \alpha_2, \ldots, \alpha_k$ to be zero.

2.2.2 Linear Span

Definition 2.5

The linear span of a set of vectors $v_1, v_2, \ldots, v_k$ in $\mathbb{R}^n$ is the set of all linear combinations of these vectors, i.e.,

$$\operatorname{span}\{v_1, v_2, \ldots, v_k\} = \{\alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_k v_k \mid \alpha_1, \alpha_2, \ldots, \alpha_k \in \mathbb{R}\}.$$

Notation

The linear span of vectors $v_1, v_2, \ldots, v_k$ is denoted by $\operatorname{span}\{v_1, v_2, \ldots, v_k\}$. Or simply by $\langle v_1, v_2, \ldots, v_k \rangle$.

In some sense the linear span is a technique to gerente a set of vectors from some (a list of) vectors.

2.2.3 Spanning Sets

Definition 2.6

A set of vectors $v_1, v_2, \ldots, v_k$ in $\mathbb{R}^n$ is called a spanning set of a vector space V if

$$V = \operatorname{span}\{v_1, v_2, \ldots, v_k\}.$$

Example

- The standard basis vectors $v_1 = (1, -1, 0), v_2 = (0, 1, -1)$ in $\mathbb{R}^3$ form a spanning set of a plane in $\mathbb{R}^3$.
$$x_1 + x_2 + x_3 = 0 \subset \mathbb{R}^3$$
- Let $e_1, e_2, \ldots, e_n$ be the standard basis vectors in $\mathbb{R}^n$. Then $\{e_1, e_2, \ldots, e_n\}$ forms a spanning set of $\mathbb{R}^n$, i.e.,
$$\mathbb{R}^n = \operatorname{span}\{e_1, e_2, \ldots, e_n\}.$$

Remark

If $\{v_1, v_2, \ldots, v_k\}$ are contained in $\operatorname{span}\{u_1, u_2, \ldots, u_m\}$, i.e.,

$$v_i \in \operatorname{span}\{u_1, u_2, \ldots, u_m\} \quad \text{for all } i = 1, 2, \ldots, k,$$

then

$$\operatorname{span}\{v_1, v_2, \ldots, v_k\} \subseteq \operatorname{span}\{u_1, u_2, \ldots, u_m\}.$$

2.2.4 Column Space and Row Space

Let A be an $m \times n$ matrix. Then its column are vectors in $\mathbb{R}^m$.

Definition 2.7

The column space of A, is the linear span of its column vectors.

$$\text{Col}(A) = \{y \in \mathbb{R}^m \mid y = Ax \text{ for some } x \in \mathbb{R}^n\}.$$

Definition 2.8

Let A be an $m \times n$ matrix. The row space of A, is the linear span of its row vectors.

$$\text{Row}(A) = \{y \in \mathbb{R}^n \mid y = xA \text{ for some } x \in \mathbb{R}^m\}.$$

2.3 Linear Independence and Dimension

2.3.1 Linearly Dependence

Definition 2.9

A set of vectors $\{v_1, v_2, \ldots, v_k\}$ in a vector space V is said to be linearly dependent if there exist scalars $\lambda_1, \lambda_2, \ldots, \lambda_k$, not all zero, such that

$$\lambda_1 v_1 + \lambda_2 v_2 + \cdots + \lambda_k v_k = 0.$$

If $\lambda_m \neq 0$, then

$$v_m = -\frac{\lambda_1}{\lambda_m} v_1 - \frac{\lambda_2}{\lambda_m} v_2 - \cdots - \frac{\lambda_{m-1}}{\lambda_m} v_{m-1}.$$

2.3.2 Linearly Independence

Definition 2.10

A set of vectors $\{v_1, v_2, \ldots, v_k\}$ in a vector space V is said to be linearly independent if it is not linearly dependent.

$$\lambda_1 v_1 + \lambda_2 v_2 + \cdots + \lambda_k v_k = 0,$$

where the only solution is $\lambda_1 = \lambda_2 = \cdots = \lambda_k = 0$.

Example

$e_1, e_2, \ldots, e_n \in \mathbb{R}^n$ are linearly independent. To see this, consider the equation

$$\lambda_1 e_1 + \lambda_2 e_2 + \cdots + \lambda_n e_n = 0.$$

Since $e_1, e_2, \ldots, e_n$ are the standard basis vectors, the only solution is $\lambda_1 = \lambda_2 = \cdots = \lambda_n = 0$.

Let a,b,c,d be some real numbers, and let

$$v_1 = \begin{pmatrix} a \\ b \end{pmatrix}, \quad v_2 = \begin{pmatrix} c \\ d \end{pmatrix}.$$

Then v_1 and v_2 are linearly independent if and only if the determinant of the matrix formed by placing v_1 and v_2 as columns is non-zero

$$\det \begin{pmatrix} a & c \\ b & d \end{pmatrix} = ad - bc \neq 0.$$

2.3.3 Linear Independence versus Linear Combination

Theorem 2.1

Let $v_1, v_2, \ldots, v_k$ be vectors in $\mathbb{R}^n$. The following statements are equivalent

- $v_1, v_2, \ldots, v_k$ are linearly independent.
- None of the vectors $v_1, v_2, \ldots, v_k$ can be written as a linear combination of the others.
- for every $x \in span\{v_1, v_2, \ldots, v_k\}$, there exists a unique set of scalars $\lambda_1, \lambda_2, \ldots, \lambda_k$ such that
$$x = \sum_{i=1}^{k} \lambda_i v_i.$$

Proof. consider x and x' in $\text{span}\{v_1, v_2, \ldots, v_k\}$ such that

$$x = \sum_{i=1}^{k} \lambda_i v_i \quad \text{and} \quad x' = \sum_{i=1}^{k} \lambda_i' v_i.$$

If $x = x'$, then

$$\sum_{i=1}^{k} \lambda_i v_i = \sum_{i=1}^{k} \lambda_i' v_i \implies \sum_{i=1}^{k} (\lambda_i - \lambda_i') v_i = 0.$$

Since $v_1, v_2, \ldots, v_k$ are linearly independent, it follows that

$$\lambda_i - \lambda_i' = 0$$

for all i, i.e., $\lambda_i = \lambda_i'$ for all i. This proves the uniqueness of the representation. □

Corollary 2.2

Suppose $v_1, v_2, \ldots, v_k$ are linearly independent. Then any sublist of $v_1, v_2, \ldots, v_k$ is also linearly independent.

2.3.4 Steinitz Exchange Lemma

Lemma 2.3

Steinitz exchange lemma

- Let V be a vector subspace of $\mathbb{R}^n$.
- Let $\{v_1, v_2, \ldots, v_m\}$ be a linearly independent set in V and let $v \in V$.
- Let $\{w_1, w_2, \ldots, w_k\}$ be a set spanning V.

One has

$$m \leq k.$$

Corollary 2.4

Any $k \geq n$ vectors in $\mathbb{R}^n$ are linearly dependent.

Example

Let M be an $n \times n$ matrix. There exists a polynomial

$$p(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0,$$

Such that

$$p(M) = M^n + a_{n-1}M^{n-1} + \cdots + a_1 M + a_0 I = 0.$$

2.3.5 Basis of a linear subspace

Definition 2.11

A basis of a linear subspace V of $\mathbb{R}^n$ is a linearly independent set of vectors in V that spans V.

$$V = \text{span}\{v_1, v_2, \ldots, v_k\}.$$

Theorem 2.5

Any two bases of a finite-dimensional vector space have the same number of elements.

Proof. We have two bases of a finite-dimensional vector space V.

$$\{v_1, v_2, \ldots, v_m\} \text{ and } \{w_1, w_2, \ldots, w_n\}$$

By the Steinitz exchange lemma, we have

$$m \leq n \text{ and } n \leq m.$$

Therefore,

$$m = n.$$

□

Lemma 2.6

Delection Lemma

If $v_1, v_2, \ldots, v_k$ is a linearly dependent set in a vector space V and

$$V \subseteq \text{span}\{v_1, v_2, \ldots, v_k\},$$

We can find an index i such that v_i can be expressed as a linear combination of the other vectors in the set.

$$v_i \in \text{span}\{v_1, v_2, \ldots, v_{i-1}, v_{i+1}, \ldots, v_k\}.$$

v_i is called a deleted vector.
Then we can remove v_i from the set without changing the span of the set.

Lemma 2.7

Extension Lemma

If $v_1, v_2, \ldots, v_k$ is a linearly independent set in a vector space V and

$$\text{span}\{v_1, v_2, \ldots, v_k\} \subseteq V,$$

then there exists an index i such that v_i can be expressed as span of the other vectors in the set.

$$v_i \in \text{span}\{v_1, v_2, \ldots, v_{i-1}, v_{i+1}, \ldots, v_k\}.$$

2.3.6 Dimension

Definition 2.12

The dimension of a vector space V is the number of vectors in any basis of V.

Lemma 2.8

Let W be a linear subspace of $\mathbb{R}^n$ such that $V \subseteq W$. Then

$$\dim(V) \leq \dim(W).$$

Moreover, if $\dim(V) = \dim(W)$, then $V = W$.

2.4 Arbitrary Field $\mathbb{F}$

Let $\mathbb{F}$ be an arbitrary field, Everything we did so far could be done in $\mathbb{F}^n$ instead of $\mathbb{R}^n$.

Theorem 2.9

If $\mathbb{F}$ is a finite field, then

$$|\mathbb{F}| = p^n.$$

for some prime number p and positive integer n.

Proof. Since $\mathbb{F}$ is a finite field

$$1, 1+1, 1+1+1, \ldots$$

Since $\mathbb{F}$ is finite, this sequence must eventually repeat, implying that there exists a positive integer p such that

$$\underbrace{1+1+\cdots+1}_{p \text{ times}} = 0.$$

So since p is the smallest positive integer such that the sum of 1 repeated p times equals 0, we have

$$\mathbb{F}_p \subseteq \mathbb{F}.$$

Then $\mathbb{F}$ can be viewed as a vector space over $\mathbb{F}_p$. Let $n = \dim_{\mathbb{F}_p}(\mathbb{F})$. Then

$$|\mathbb{F}| = p^n.$$

□

2.5 Orthogonality

2.5.1 Orthogonal Vectors

Definition 2.13

Vectors $v_1, v_2, \ldots, v_k$ in a vector space V are called orthogonal if

$$\langle v_i, v_j \rangle = 0 \quad \text{for all } i \neq j.$$

We say that these vectors are orthonormal if

$$v_i \cdot v_j = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

Lemma 2.10

If $v_1, v_2, \ldots, v_k$ are orthogonal vectors in a vector space V, then they are linearly independent.

Corollary 2.11

Orthonormal vectors in a vector space V are linearly independent.

2.5.2 Orthonormal Basis

Definition 2.14

Orthonormal basis of a vector space V is a basis consisting of orthonormal vectors.

Let x be a vector in a vector space V with an orthonormal basis $\{v_1, v_2, \ldots, v_n\}$. Then x can be expressed as

$$x = \sum_{i=1}^{n} \langle x, v_i \rangle v_i.$$

Much easier to compute the dot product with the orthonormal basis.

2.5.3 Orthonormal Complement

Definition 2.15

Let V be a vector space with an inner product, and let W be a subspace of V. The orthonormal complement of W in V, denoted by $W^\perp$, is the set of all vectors in V that are orthogonal to every vector in W:

$$W^\perp = \{v \in V \mid \langle v, w\rangle = 0 \text{ for all } w \in W\}.$$

The subset $W^\perp$ is itself a subspace of V.

Lemma 2.12

Let $v_1, v_2, \ldots, v_m$ be vectors in V that span V Then

$$V^\perp = \{v \in V \mid \langle v, v_i\rangle = 0 \text{ for all } i = 1, 2, \ldots, m\}.$$

2.5.4 Orthogonal Decomposition

Theorem 2.13

Let V be a linear subspace of $\mathbb{R}^n$. Then

1. $V \cap V^\perp = \{0\}$
2. for each $x \in \mathbb{R}^n$, one has
$$x = v + w,$$
for unique $v \in V$ and $w \in V^\perp$.
3. $\dim V + \dim V^\perp = n$

Idea of proof: let $v = P(x)$ and $w = x - P(x)$ for

$$P(x) = \sum_{i=1}^{m} \langle x, v_i\rangle v_i.$$

Then $v \in V$ and $w \in V^\perp$, and the decomposition is unique.

2.5.5 Orthogonal Projection

Definition 2.16

Let V be a linear subspace of $\mathbb{R}^n$, and let $x \in \mathbb{R}^n$. The orthogonal projection of x onto V, denoted by $P_V(x)$, is the vector $v \in V$ such that

$$x - v \in V^\perp.$$

If $\{v_1, v_2, \ldots, v_m\}$ is an orthonormal basis of V, then

$$P_V(x) = \sum_{i=1}^{m} \langle x, v_i\rangle v_i.$$

2.5.6 Orthonormal Extension

Lemma 2.14

There is some orthonormal basis of V that contains the given orthonormal set $\{v_1, v_2, \ldots, v_m\}$.

Proof. As $\{v_1, \ldots, v_m\}$ is orthonormal, it is linearly independent. Extend it to a basis of V.

$$\{v_1, \ldots, v_m, u_1, \ldots, u_k\}$$

Applying the Gram-Schmidt process to this basis leaves the first m vectors unchanged, and orthogonalizes the remaining vectors to yield an orthonormal set

$$\{e_{m+1}, \ldots, e_{m+k}\}$$

Therefore

$$\{v_1, \ldots, v_m, e_{m+1}, \ldots, e_{m+k}\}$$

forms an orthonormal basis for V containing the given set. □

2.5.7 The Gram-Schmidt Process

Theorem 2.15

Gram-Schmidt Process

Let $\{u_1, u_2, \ldots, u_m\}$ be a linearly independent set of vectors in $\mathbb{R}^n$.
The *Gram-Schmidt process* constructs an orthonormal set $\{v_1, v_2, \ldots, v_m\}$ as follows

$$w_1 = u_1, \qquad v_1 = \frac{w_1}{\|w_1\|},$$

$$w_j = u_j - \sum_{i=1}^{j-1} \langle u_j, v_i \rangle v_i, \quad v_j = \frac{w_j}{\|w_j\|}, \quad j = 2, \ldots, m.$$

Then $\{v_1, v_2, \ldots, v_m\}$ is an orthonormal set that spans the same subspace as $\{u_1, u_2, \ldots, u_m\}$.

The step of gram-schmidt process involves subtracting the projections of the current vector onto the previously obtained orthonormal vectors and then normalizing the result.

1. $w_1 = u_1$
2. $w_2 = u_2 - \langle u_2, v_1 \rangle v_1$
3. $w_3 = u_3 - \langle u_3, v_1 \rangle v_1 - \langle u_3, v_2 \rangle v_2$
4. ...
5. $w_m = u_m - \sum_{i=1}^{m-1} \langle u_m, v_i \rangle v_i$
6. Then we normalize each w_i to obtain

$$v_i = \frac{w_i}{\|w_i\|} \qquad \text{for } i = 1, 2, \ldots, m$$